

Factory-to-Customer Shipping Route Efficiency Analysis

Nassau Candy Distributor

AUTHOR	ROLE	DATE	TOOLS
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Abstract

Nassau Candy ships from five factories to customers in 49 states plus Washington, D.C., but had no real way of knowing which of those routes actually worked well. This paper walks through how I answered that: cleaning and validating 9,994 order records (8,389 unique orders), running into a serious problem with the shipping date data early on, and working through what that meant for everything built afterward. The short version of that problem is that Shipping Lead Time — the number nearly every KPI in this project depends on — doesn't reflect real shipping chronology. Rather than throw it out, I treated it as a relative measure instead of a literal one, and built every route, factory, and ship mode comparison on that basis. What came out of it is a set of findings I'm confident stand up to scrutiny: a bottleneck in Washington state that shows up from two separate factories, and a ship mode structure with no real cost-speed trade-off at all. All of it is backed by SQL analysis, a pandas KPI layer, and an interactive Streamlit dashboard.

1. Introduction & Background

Nassau Candy ships confectionery products from five factories to customers across the United States, and like most distribution businesses, how well that shipping actually runs matters — to customer satisfaction, to operating cost, to how well the business can grow. The company had plenty of order and shipment data sitting around, but no structured way to actually use it to answer basic questions: which routes are reliable, which aren't, and where problems tend to cluster geographically. Without that visibility, logistics decisions were being made reactively rather than from evidence.

2. Problem Statement

I scoped this analysis around four questions: which factory-to-customer routes are consistently efficient, which ones see frequent delays, how performance differs by region, state, and ship mode, and where the geographic bottlenecks actually are. Throughout, the goal wasn't to produce a pile of descriptive statistics — it was to end up with specific, named routes and states an operations team could actually do something about.

3. Dataset Overview

The raw dataset contained 10,194 order line items across 18 fields, including Order Date, Ship Date, Ship Mode, Product Name, Division, and full destination geography (City, State/Province, Region, Postal Code). A separate reference table provided factory locations (with coordinates) and a Product-to-Factory mapping across five factories: Lot's O' Nuts, Wicked Choccy's, Sugar Shack, Secret Factory, and The Other Factory. After cleaning (detailed in Section 4), the working dataset totals 9,994 U.S. order line items across 8,389 unique orders and 5,009 unique customers.

4. Methodology

4.1 Data Cleaning & Validation

Two cleaning decisions shaped the working dataset. First, 200 records (about 2% of the raw data) belonged to Canadian provinces; since the project is scoped to U.S. distribution and the dashboard requires a U.S. state-level map, these were removed. Second, the raw Division field labeled one product — Fizzy Lifting Drinks — as "Sugar," while both its Product ID prefix and the official Product–Factory mapping table indicated "Other." Since two of the three sources agreed, the Factory join was built from Product Name rather than the unreliable Division field. A full null check across all 24 working columns returned zero missing values.

4.2 Data Quality Investigation: The Shipping Lead Time Anomaly

Shipping Lead Time — defined as Ship Date minus Order Date — is the metric the majority of this project's KPIs depend on, so it received close scrutiny before anything was built on top of it. The raw values ranged from 904 to 1,642 days, immediately implausible for a distribution business. Four independent checks confirmed the problem was structural, not a handful of outliers:

- **Date range mismatch:** Order Date spans only 2024–2025, while Ship Date spans 2026–2030 — the two fields do not move together the way a real order-to-shipment relationship would.
- **Non-continuous distribution:** a histogram of Lead Time shows three sharp, isolated spikes (around 910, 1,275, and 1,640 days) with no values in between, rather than the smooth spread a genuine shipping delay would produce.
- **Year-gap sub-groups:** grouping by the gap between Ship Date year and Order Date year reveals four distinct sub-groups. Two are extremely tight (standard deviation of 1–2 days), and two are wide and overlapping (standard deviation of 166–181 days) — this overlap is what causes the histogram to show three visual clusters rather than four.
- **Year-normalization test:** forcing both dates to a common year (removing the year mismatch entirely and comparing only month and day) still produced a spread centered near zero, with roughly half the resulting values negative. This rules out a simple "wrong year label" explanation — the month/day components do not preserve a real chronological relationship either.

A supporting observation surfaced later, during Ship Mode analysis: Same Day and First Class shipments both average a *longer* Lead Time than Standard Class — the reverse of what these category names should mean in practice, and consistent with a dataset where the date fields were not generated to reflect real shipping behavior.

Resulting methodology: Lead Time is calculated and used throughout this project — it underpins the KPIs the dashboard and this paper both depend on — but only as a **relative, comparative** measure: this route versus that one, this ship mode versus another, never as a literal "customers wait X days" claim. This decision was raised for mentor guidance before being finalized; the guidance received was that the dataset should be used as given and the problem statement followed closely, which was read as direction to work around the limitation with sound judgment rather than search further for a technical fix. This finding is specific to the two date fields — it was not observed in, and should not be assumed to apply to, other fields such as Sales, Cost, or geography.

4.3 Feature Engineering

Beyond Lead Time, two route-identifying fields were built to support the KPIs in Section 5: Route_State (Factory → customer State/Province, e.g., "Wicked Choccy's → Texas") and Route_Region (Factory → customer Region), producing 177 and 20 unique combinations respectively. A Route Efficiency Score was also engineered — average Lead Time per route, min-max normalized to a 0–100 scale where 100 marks the fastest qualifying route and 0 the slowest, restricted to routes with at least 10 orders to avoid low-volume statistical noise. A Delay Frequency metric was defined similarly: the share of orders whose Lead Time falls at or above 1,634 days, the boundary of the

slowest of the three identified clusters described in Section 4.2.

4.4 Analytical Approach

Analysis proceeded in four stages: exploratory cleaning and the Lead Time investigation in Python (pandas), a 27-query SQL analysis in MySQL covering data validation, overall KPIs, route and factory performance, delay frequency, efficiency scoring, and cost-time comparisons, a pandas rebuild of the same KPI logic (necessary because the Streamlit deployment target cannot connect to a local MySQL instance), and finally the interactive Streamlit dashboard itself. Every KPI computed in SQL was cross-checked against its pandas equivalent and confirmed identical before being used in this paper or the dashboard.

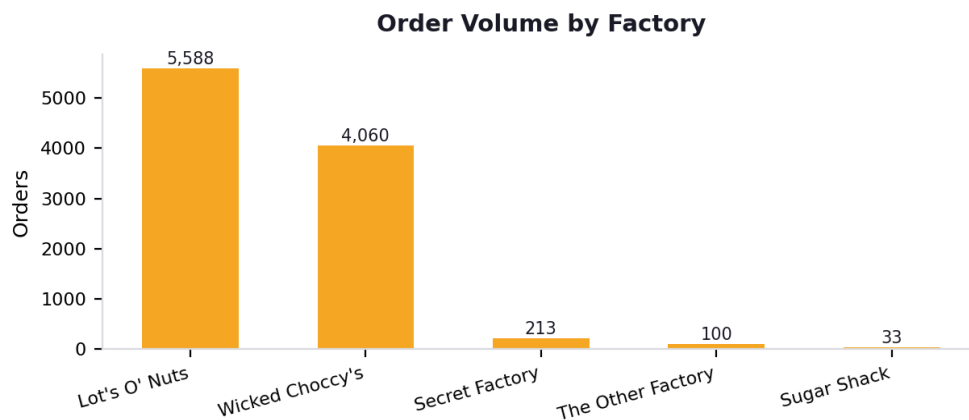
5. Exploratory Data Analysis & Key Findings

5.1 Overall Metrics

Metric	Value
Total Orders (unique)	8,389
Total Sales	\$138,830.34
Total Gross Profit	\$91,508.23
Overall Delay Frequency	33.26%
Average Route Efficiency Score	54.7 / 100

5.2 Factory Performance

Order volume is heavily concentrated: Lot's O' Nuts (5,588 orders) and Wicked Choccy's (4,060 orders) together account for roughly 96.5% of all shipments. This reflects the small number of high-volume chocolate products mapped to these two factories rather than any operational or data issue. Secret Factory, The Other Factory, and Sugar Shack handle the remaining 213, 100, and 33 orders respectively.



5.3 Route Performance

Across 177 qualifying state-level routes, Wicked Choccy's → Nevada ranks as the most efficient (Efficiency Score 100, 10 orders), and Wicked Choccy's → New Mexico the least (Efficiency Score 0, 14 orders). Ranking alone, however, does not account for operational significance — a slow route with 10 orders matters far less than one with hundreds.

Rank	Route	Volume	Efficiency Score
1 (fastest)	Wicked Choccy's → Nevada	10	100.0
2	Secret Factory → Texas	16	85.4
3	Lot's O' Nuts → Virginia	91	84.6
4	Wicked Choccy's → South Carolina	14	83.1
5	Lot's O' Nuts → Utah	21	82.5
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173	Lot's O' Nuts → Connecticut	40	22.3
174	Lot's O' Nuts → New Mexico	18	10.4
175	Secret Factory → Washington	11	5.4
176	Lot's O' Nuts → Iowa	14	3.2
177 (slowest)	Wicked Choccy's → New Mexico	14	0.0

Table 1. Top 5 and bottom 5 state-level routes by Route Efficiency Score.

Filtering to routes with both above-average volume and above-average Lead Time surfaces six routes, and Washington appears twice: Lot's O' Nuts → Washington (230 orders, Efficiency Score 47.3) and Wicked Choccy's → Washington (187 orders, Efficiency Score 36.9 — the weakest of the six). That two separate factories both underperform serving the same state is a stronger signal than any single-route outlier. The remaining flagged routes are Lot's O' Nuts → New York (544 orders), Lot's O' Nuts → Ohio (218 orders), Wicked Choccy's → Pennsylvania (188 orders), and Wicked Choccy's → Illinois (183 orders).

5.4 Geographic Bottlenecks

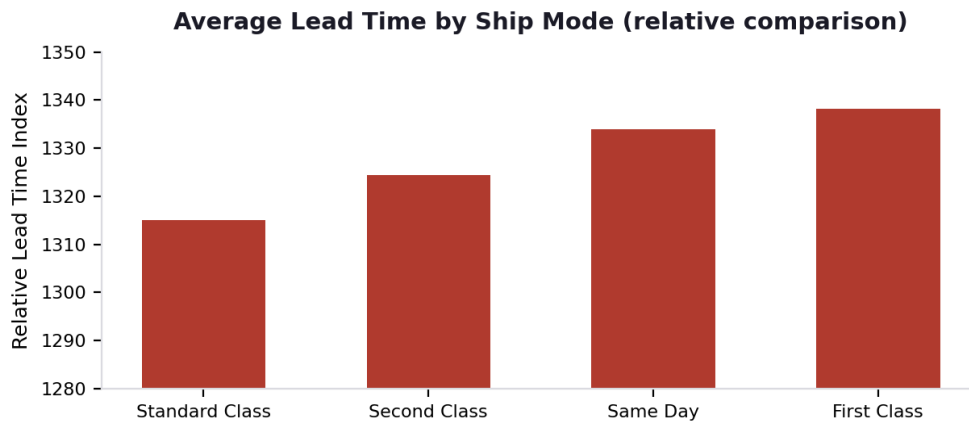
At the state level (aggregating across factories), Illinois, Washington, and Ohio combine above-average order volume with above-average Delay Frequency. Washington is the worst of the three at 42.82% (433 and 392 orders for Illinois and Ohio, at 35.57% and 35.46% respectively). California is a notable exception: despite the highest order volume of any state (1,686 orders), its delay rate (32.92%) sits just under the cross-state average — an average pulled upward by smaller states with more erratic performance. Volume and delay rate do not move together uniformly across this dataset.

5.5 Ship Mode Performance

Standard Class accounts for the majority of orders (5,026 of 8,389), ahead of Second Class (1,621), First Class (1,302), and Same Day (440). Cost per shipment is narrow across all four modes (roughly \$4.42–\$4.84), so there is no meaningful cost-time trade-off to report — faster-sounding modes are not meaningfully more expensive in this data.

Ship Mode	Orders	Avg. Cost/Shipment	Relative Lead Time Rank
Standard Class	5,026	\$4.73	1 (fastest)
Second Class	1,621	\$4.84	2
Same Day	440	\$4.42	3
First Class	1,302	\$4.73	4 (slowest)

Table 2. Ship Mode summary. Rank reflects relative Lead Time comparison only, not literal shipping days (see Section 4.2).



As noted in Section 4.2, both First Class and Same Day average a longer Lead Time than Standard Class, a pattern that runs counter to normal logistics expectations and further supports the data quality finding.

6. Discussion

Two things from this analysis feel like they actually matter operationally, not just statistically. The first is Washington — it's not a one-off outlier, it shows up as a problem three separate ways: in the high-volume/poor-performance route analysis, in the state-level congestion analysis, and as the single highest state-level delay rate in the whole dataset. Since it's underperforming from both major factories at once, I'd guess the cause sits with the destination side — regional logistics or carrier handling in that corridor — rather than anything specific to either factory. The second is the ship mode result: there's simply no cost-speed trade-off here. That's worth re-checking once the date data is fixed, since it's entirely possible the real picture looks different with trustworthy Lead Time numbers.

7. Limitations

- The central limitation is the Order Date/Ship Date data quality issue detailed in Section 4.2. Every Lead-Time-derived metric in this paper — Efficiency Score, Delay Frequency, and the ship mode comparison — is valid only as a relative measure, not as an absolute, real-world time figure.
- The Delay Frequency threshold (Lead Time $\geq 1,634$) is derived from the dataset's own cluster structure rather than an externally defined service-level target, since no such target exists given the underlying data issue.
- This investigation focused specifically on the date fields; other fields (Sales, Cost, Units) were not subjected to the same depth of scrutiny and should not be assumed either reliable or unreliable without separate validation.
- State-level Delay Frequency comparisons use an unweighted average across states, which can be skewed by small, high-variance states — a factor already visible in the California result in Section 5.4.

8. Recommendations

- **Prioritize Washington-bound shipments for review.** It is the only state flagged as a bottleneck from both major factories simultaneously, and it has the highest state-level delay rate in the dataset.
- **Extend the same review to Illinois and Ohio**, which show the same above-average volume/above-average delay pattern at a less extreme level.
- **Do not use Ship Mode as a cost-speed lever** until the underlying data issue is resolved — the current data shows no trade-off to exploit, and category names do not currently reflect actual relative speed.
- **Resolve the Order Date/Ship Date generation issue at the source.** This is the single change that would let Lead Time be used as a trustworthy, absolute metric rather than requiring the relative-only treatment applied

throughout this analysis.

- **Re-run this analysis once the date issue is fixed** to confirm whether the route, geographic, and ship mode findings above hold under genuine shipping chronology, or shift once trustworthy absolute Lead Time values are available.

9. Conclusion

I set out to give Nassau Candy the route-level visibility it didn't have, and I think this delivers that — with one caveat I've tried to be upfront about throughout: the Order Date and Ship Date fields don't reflect real shipping chronology, so every Lead-Time-based finding here is a relative comparison, not a literal timing claim. Within that constraint, there are patterns here worth acting on: a Washington bottleneck that shows up from two separate factories, smaller but real concerns in Illinois and Ohio, and a ship mode structure that shows no actual cost-time trade-off. Everything is cross-checked between SQL and pandas, and it's all explorable directly in the accompanying Streamlit dashboard. If nothing else, this should move Nassau Candy's shipping decisions from reactive to actually data-driven — and it points squarely at where I'd start once the underlying date issue gets fixed.